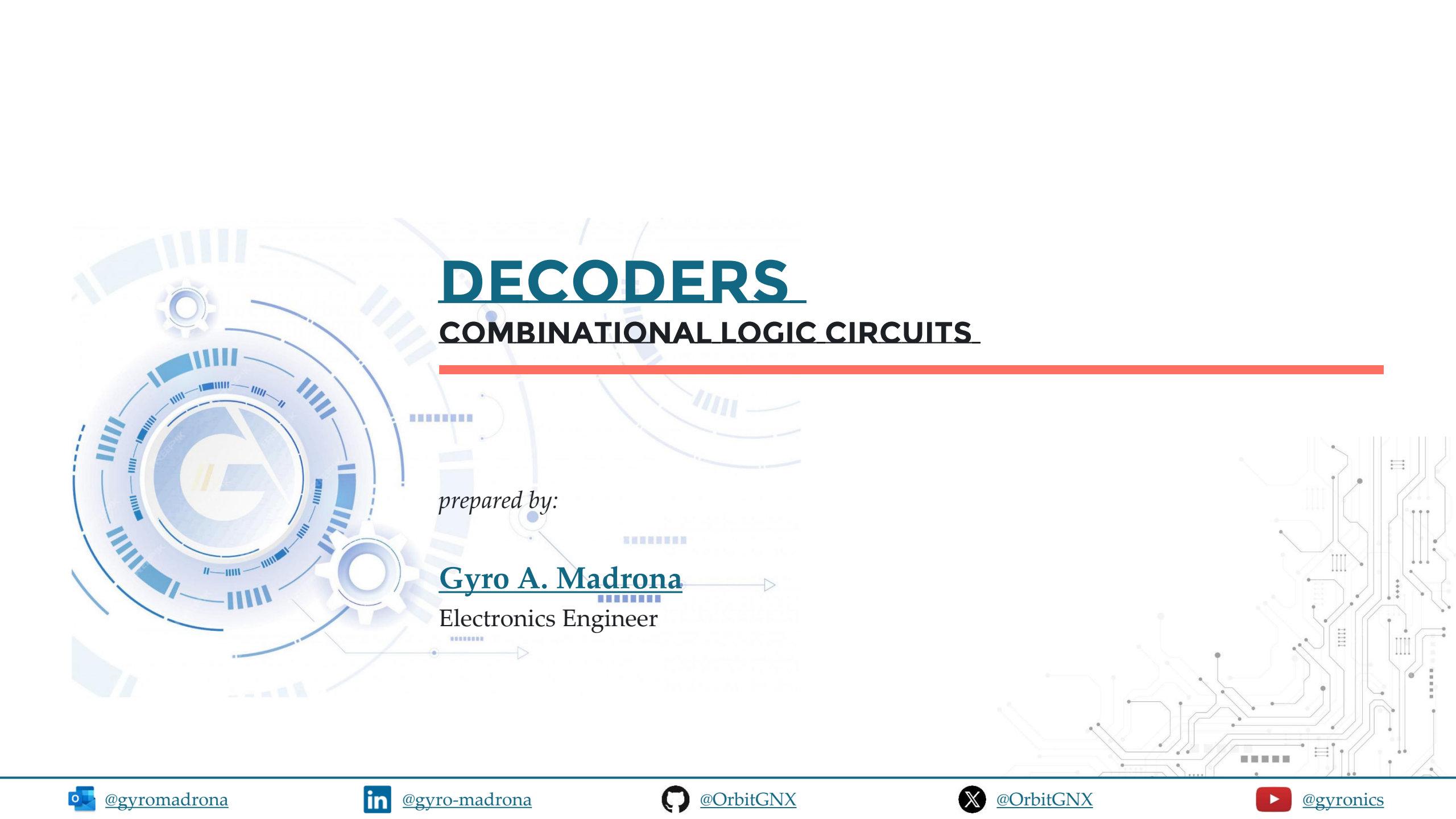




DECODERS

COMBINATIONAL LOGIC CIRCUITS

prepared by:

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TOPIC OUTLINE

Binary Decoder

BCD-to-7-Segment Decoder

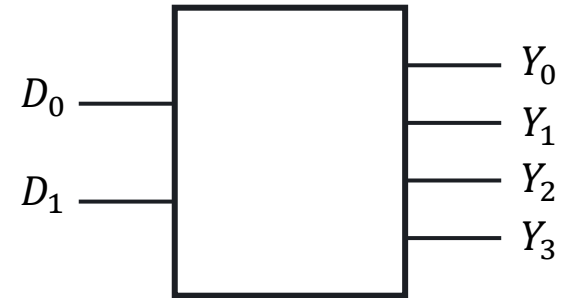


BINARY_DECODER



BINARY DECODER

Graphical Symbol



A binary decoder takes n input bits and activates one of 2^n output lines.

Truth Table

D_1	D_0	Y_0	Y_1	Y_2	Y_3
0	0	1	0	0	0
0	1	0	1	0	0
1	0	0	0	1	0
1	1	0	0	0	1

EXERCISE

Given the truth table, synthesize a 2-to-4 binary decoder with an enable input.

Solution

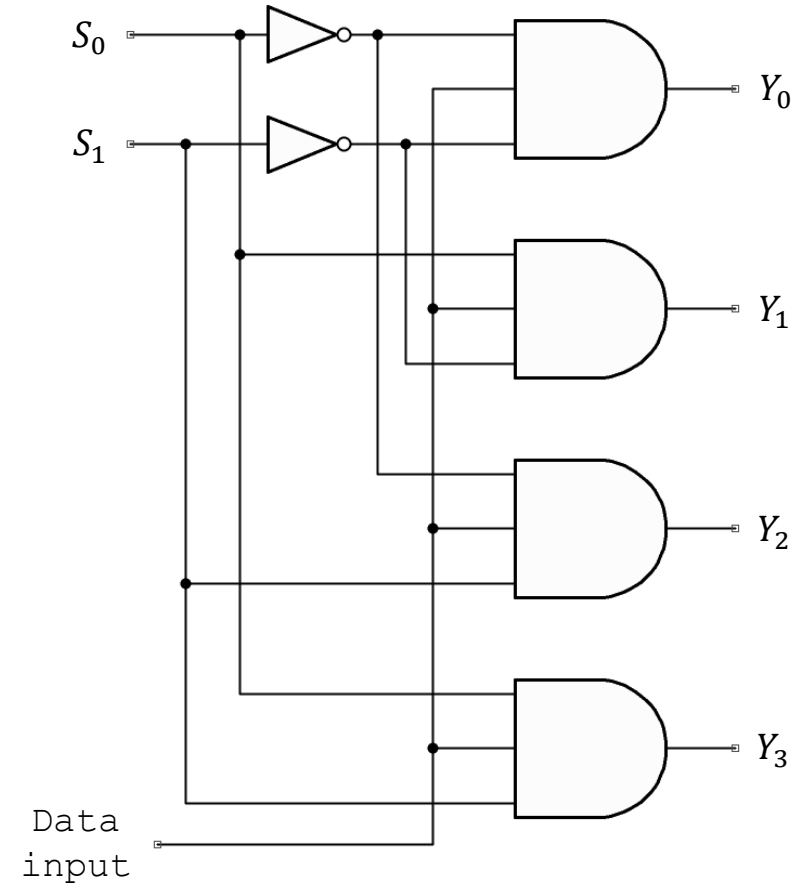
En	D_1	D_0	Y_0	Y_1	Y_2	Y_3
1	0	0	1	0	0	0
1	0	1	0	1	0	0
1	1	0	0	0	1	0
1	1	1	0	0	0	1
0	x	x	0	0	0	0



DEMULTIPLEXER

A demultiplexer is a digital circuit that routes a single data input to one of several outputs based on select input lines, performing the inverse function of a multiplexer.

A 1-line-to-4-line Demultiplexer



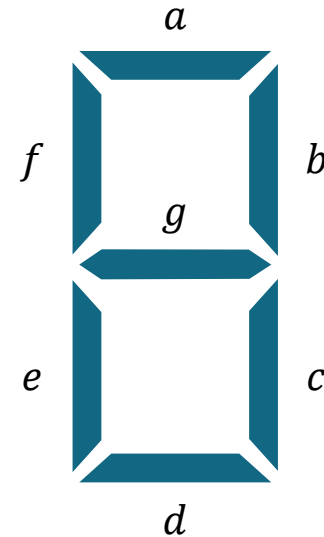
BCD-TO-7-SEGMENT DECODER



THE 7-SEGMENT DISPLAY

A standard 7-segment display consists of seven LEDs (segments) arranged in a rectangular layout to form the number 8. Each segment is labeled from *a* to *g*, and an optional eighth segment (DP) is used for the decimal point.

Segment Arrangement



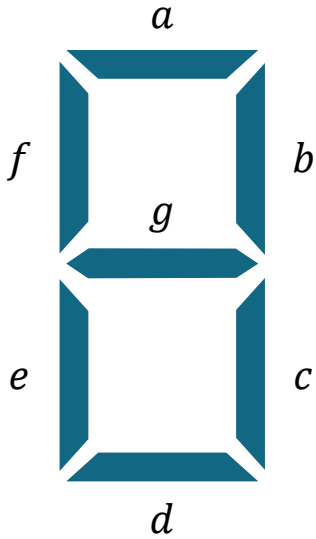
EXPRESSION FOR SEGMENT A

Truth Table

N	$DCBA$	f_a
0		
1		
2		
3		
4		
5		
6		
7		
8		

N	$DCBA$	f_a
9		
10		
11		
12		
13		
14		
15		

Segment Arrangement



EXPRESSION FOR SEGMENT A

Truth Table

N	$DCBA$	f_a
0		
1		
2		
3		
4		
5		
6		
7		
8		

K-Map

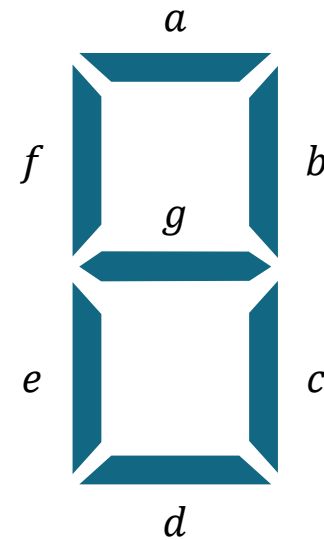
N	$DCBA$	f_a
9		
10		
11		
12		
13		
14		
15		



EXERCISE

Implement a combinational logic circuit that functions as a decoder for a 7-segment display.

Segment Arrangement



LABORATORY

